

TrayGuard: What We Learned About Why Tray Errors Persist

EC ENGR 180DW — Final Design Review

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Roadmap

- 1 The Problem
- 2 Why Tray Errors Persist
- 3 CV Attempt
- 4 Why CV Alone Won't Work
- 5 Why Training Is the Best Entry Point
- 6 What a Future Team Should Do
- 7 Conclusion

The Problem

The Problem By The Numbers

9 per 100 cases — Alfred et al.
[1] 3,900 defects in 41,799 cases
55% at assembly

10 min avg delay — Nichol et al.
[2] 236 errors, 147 cases
\$6–9M/yr estimated cost

44% wrong spec — Zhu et al.
[3] 398 errors in 33,839 packages
largest single category

These are recurring, measured operational failures — not isolated accidents.

Why Tray Errors Persist

Lookalike Instruments: Kelly vs Crile

Kelly and Crile forceps comparison



Kelly and Crile forceps: Both have transverse serrations. Used for medium to large vessels. Crile's serrations along entire length of jaws. Kelly's serrations only on 1/2 of the distal jaws

Kelly Forceps sizing



Kelly and Crile: same family, different serration patterns. Sizes 6", 7", 8" — a staff might mix 14cm and 16cm clamps. Total count correct, specification wrong. [4]

What Tray Reconstruction Actually Requires



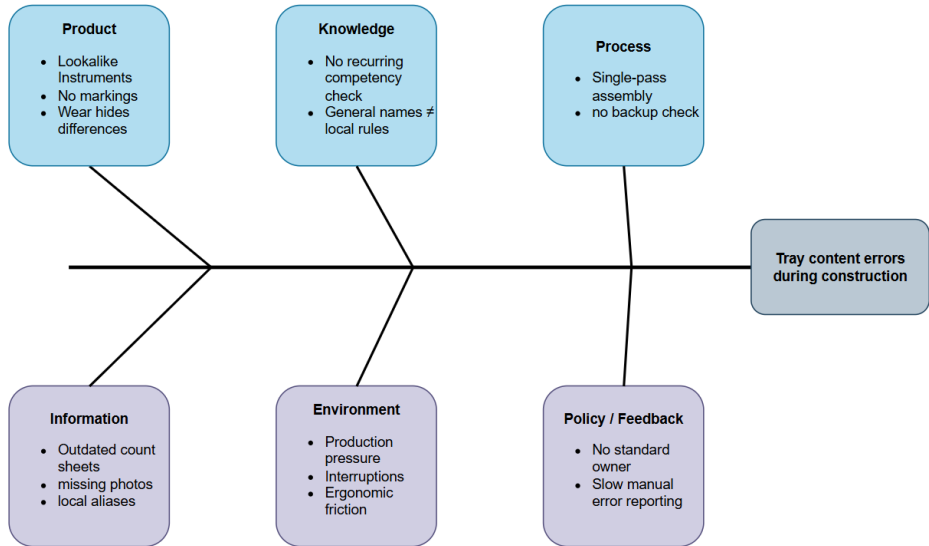
SPD assembly workstation [5]

- Identify each instrument by sight (no labels)
- Match against a count sheet with **local names**
- Distinguish lookalikes (straight vs curved Mayo)
- Check for damage, wear, contamination
- Count correct quantities
- All under **production pressure**

Common Surgical Instruments

Instrument	Manufacturer / Formal Name	Common SPD / OR Name (Local Name)
	Rochester-Ochsner Forceps (Curved, 1x2 teeth)	Kocher Clamp 
	Halsted Mosquito Forceps (Straight, 1x2 teeth)	Mosquito Clamp 
	Foerster Sponge Forceps (Serrated tips)	Sponge Stick 
	Schnidt Tonsil Forceps (Curved, serrated)	Tonsil Clamp 
	Mayo-Hegar Needle Holder (With tungsten carbide inserts)	Needle Driver 
	Backhaus Towel Clamp (For drapes and towels)	Towel Clip 
	Adson Tissue Forceps (1x2 teeth)	Rat-Tooth Forceps 
	Dressing Forceps (Smooth, straight)	Smooth Forceps 
	Metzenbaum Scissors (Curved)	Metz 
	Richardson Retractor (Double-ended)	Rich Retractor 

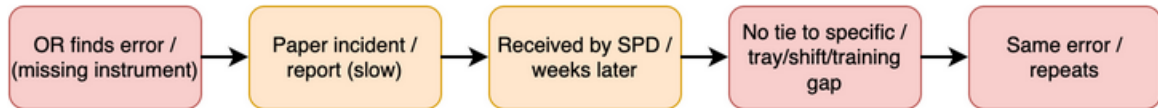
Root Cause Analysis



Root Cause Analysis

- ① SPD evolved from materials management → **no licensure, weak professionalization**
- ② Cost-center accounting → wage investment is visible, error cost is invisible
- ③ No information architecture → downstream OR errors don't trigger upstream correction

The Broken Feedback Loop



"Error reporting is cumbersome, human-dependent, delayed, and incomplete." — Nichol et al.

CV Attempt

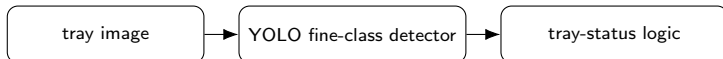
First CV Attempt: One Detector Did Everything

1-stage pipeline

- 1 Overhead image of tray
- 2 YOLO detects every fine class directly
- 3 Count detections against the tray list
- 4 Flag missing, wrong, extra, or uncertain items

Why this was brittle

- Localization and fine ID were coupled
- Similar tools shared shape and metal texture
- One wrong subtype became a tray error
- Overlay and lighting shifts reduced reliability



The Problem Was Fine-Grained Confusion, Not Just Detection

Flat fine labels

Fine class	Visual family
Scalpel n4	scalpel
Straight dissection clamp	clamp
Straight Mayo scissor	scissors
Curved Mayo scissor	scissors

What the tests showed

- Clean matched data: high mAP was achievable
- Layout shift: separated $\rightarrow$ overlay dropped sharply
- Real instruments: per-class AP depended on split
- Lookalikes needed a narrower decision context

The detector should first answer “**what family is this?**” The classifier should answer “**which subtype?**”

Second CV Attempt: Coarse-to-Fine 2-Stage Pipeline

Stage 1: general detector

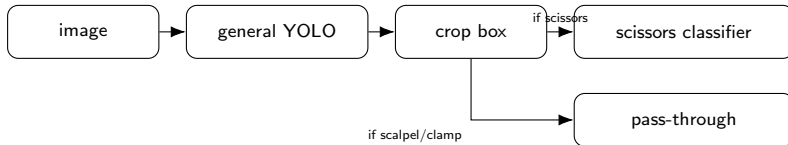
YOLO predicts broad families only:

- scalpel
- clamp
- scissors

Stage 2: specific classifier

Only runs when a family has variants:

- scissors → straight vs curved
- scalpel → pass-through
- clamp → pass-through



Testing the 2-Stage Demo on Separated Images

Dataset builder

- Remap fine labels into general classes
- Reuse ground-truth boxes for YOLO labels
- Crop boxes into ImageFolder classifier data
- Train one selected specific classifier

End-of-training

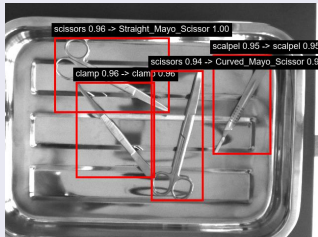
- Run on all separado*.jpg images
- Draw general boxes and confidence
- Classify cropped scissors as straight/curved
- Export annotated images, crops, CSV, HTML report

What the training shows

General detection and fine-grained scissors classification can be tested together while keeping their failure modes separate.

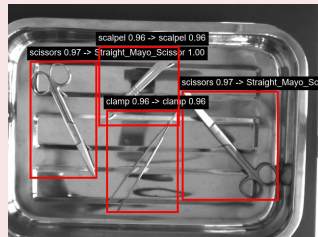
2-Stage Helped, But Scissor ID Still Failed

Helped



- Broad family detection worked.
- Scissors crops routed to classifier.

Still failed



- Both scissors predicted as Straight_Mayo.
- Fine-grained lookalike ID remained unreliable.

Takeaway

The 2-stage design separated detection from subtype classification, but it was still not reliable enough for deployment.

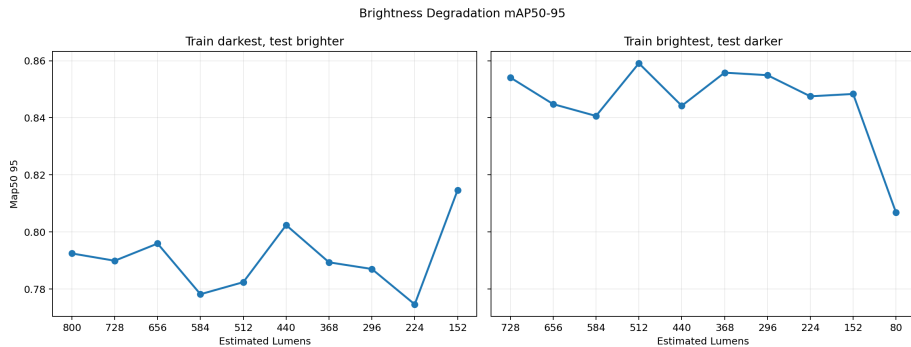
Why CV Alone Won't Work

We Ran 7 Staged Detection Experiments

Experiment	What it tested	Key result
Bright→dim	Light tolerance	mAP50-95 = 0.781
Matte→reflective	Background transfer	mAP50-95 = 0.759
Separated→overlay	Occlusion	mAP50-95 = 0.389
Real instruments	Shape vs position	Per-class AP: 0.000–0.862

Lavado baseline (clean data): **mAP50 = 0.990**. Pipeline works — the data doesn't generalize.

Brightness Degradation



Occlusion Is the Dealbreaker

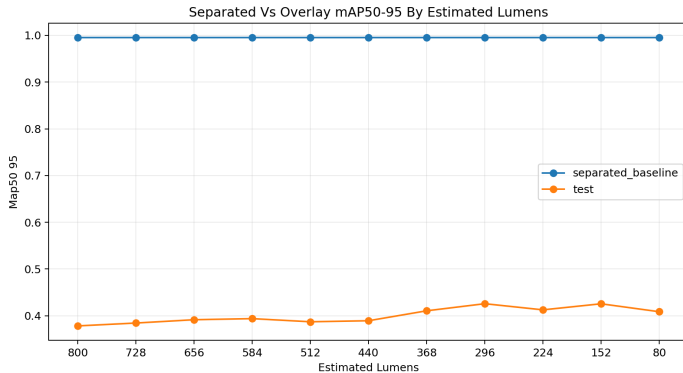
Separated layout (training)



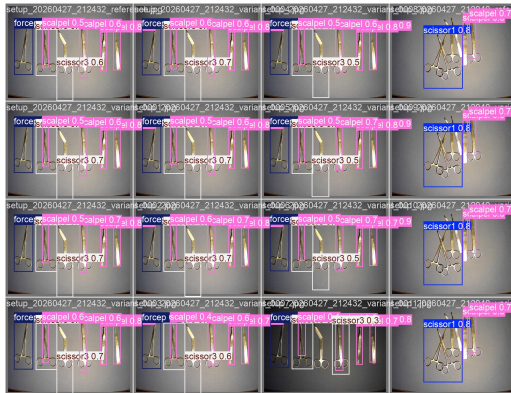
Overlay layout (test)



Overlay Degradation



Real Instruments: The Model Learned Position, Not Shape



Split	Surviving classes	Zero AP classes
Matte→reflective	scissor2, forcep, scalpel	scissor1/3/4
Reflective→matte	scalpel only	scissor1/2/3/4, forcep
Shape similarity	scissor1, scissor4	scissor2/3, forcep

The Cumulative Lesson: CV Validation Burden

What a deployment-grade CV system still needs

- ✗ Site-specific training data (instruments, lighting, layouts)
- ✗ Cross-manufacturer validation [6] (Kienle: major performance drop)
- ✗ Open-set rejection (unknown objects = false positives)
- ✗ Human-in-the-loop for every uncertain case
- ✗ Workflow integration in live SPD (never tested)
- ✗ Regulatory / device policy / privacy review

We Evaluated Seven Solution Categories

Solution	Buildable?	Validatable?	Creates infra for future?
CV tray checker	No (needs SPD)	No	Maybe
Physical error-proofing	No (custom fab)	No	No
Workflow redesign	No (policy change)	No	No
Decision support	Partial	Partial	Yes (local content)
Task simplification	No (hospital data)	No	No
Accountability system	No (org policy)	No	No
Training (selected)	Yes	Yes	Yes (module schema)

Why Training, Not Because Training Is a Cure

What training **addresses**

- Measurable local tray knowledge
- Lookalike discrimination practice
- Pre/post learning evidence

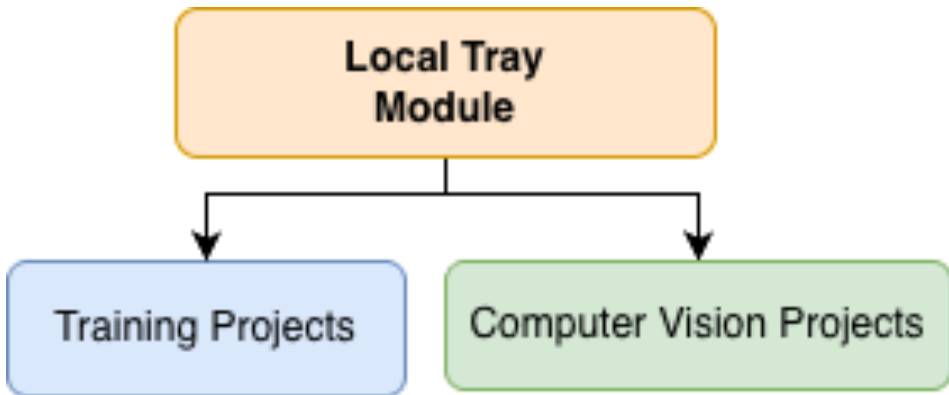
What training does **NOT** fix

- Broken count sheets
- Staffing shortages
- Structural wage problems
- The broken feedback loop

Training is the one piece we can build and validate **now**.

Why Training Is the Best Entry Point

CV and Training Are Complementary



The module is the durable artifact. The training loop proves the module works. The module then enables CV.

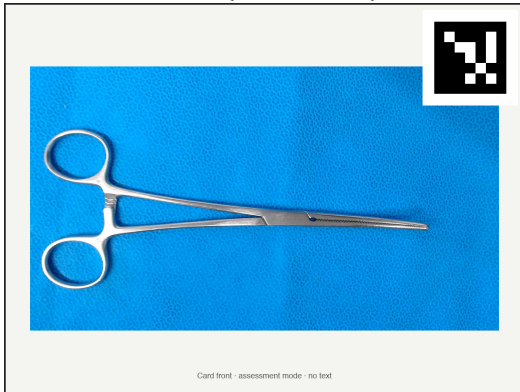
The Durable Design Artifact: Tray Module Schema

```
{
  "id": "mayo_scissors_straight_55",
  "display_name": "Straight Mayo, 5.5 in",
  "aliases": ["straight Mayo",
              "suture scissors"],
  "distinguishing_features": [
    "Heavy straight blades"
  ],
  "common_confusions": [
    "mayo_scissors_curved_55"
  ],
  "apriltag_id": 12,
  "study_prompts": [ ... ]
}
```

- Mirrors real count-sheet fields [7]
- Adds aliases, lookalikes, prompts
- File-backed, versioned
- **Software is throwaway.**
- **Schema survives.**

Printed Cards: How It Works Physically

Front (assessment)



Back (study)

Clamp Kelly 8"

Family: Clamping/occluding - Catalog: CLAMP KELLY 8"

Key identification features

- Curved jaws with ring handles
- Transverse serrations on distal half of jaws
- 8-inch overall length

Common lookalikes

- Crile clamp — serrations run full jaw (Kelly: distal half only)
- Kocher clamp — single tooth at tip (Kelly: no tooth)
- Kelly 6" / Kelly 12" — same jaw shape, different length

Also known as: JARIT_106-191

Source: FGVC12 shared Google Drive major tray image, class JARIT_106-191: CLAMP KELLY 8"

Neutral front prevents answer leakage. Tag ID maps to instrument ID.

The Scoring Architecture: 5 Error Categories

Category	Definition	Example
Correct	Right item, right count	1 scalpel handle ✓
Missing	Required item not selected	No Kelly clamp
Wrong	Item not on tray list	Retractor not in this tray
Extra	Distractor selected	Item from different tray
Misidentified	Right class, wrong variant	Curved instead of straight
Wrong count	Right item, wrong quantity	2 Kellys, need 4

- Maps to literature error categories [1, 3, 2]
- Can separate *what* improved (fewer misses? fewer lookalikes?)

Confidence Capture

Confidence: How sure are you?

Rate your confidence for each tray selection:

1	2	3	4	5
Guessing	Unsure	Neutral	Fairly sure	Certain

Not Sure

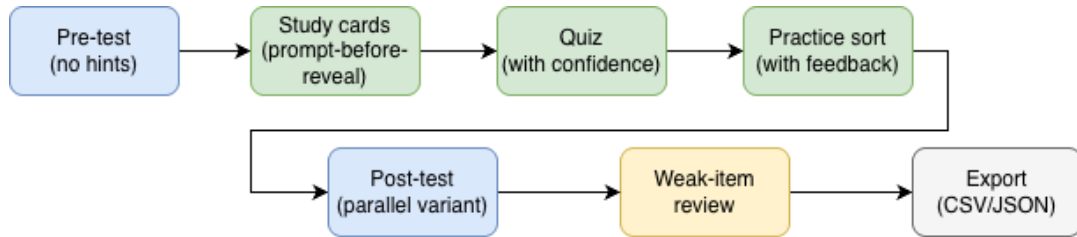
High-confidence error

Learner was sure but wrong. — Kruger & Dunning [8]

Low-confidence correct

Learner was unsure but right. — Dunlosky et al. [9]

The Learning Loop (7 Steps)



Pre / post use parallel variants. Study cards are retrieval practice. Practice gives error-specific feedback.

Pre-test (baseline) → Study cards → Quiz → Practice sort → Post-test → Export

The Research Question

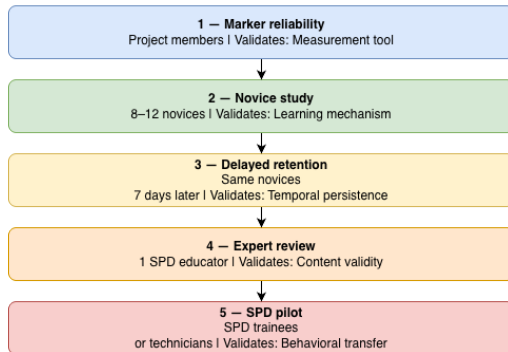
How can a local tray training and assessment system help novice users improve their ability to detect missing, wrong, extra, misidentified, or miscounted instruments before supervised SPD tray-reconstruction work?

That question is narrow, testable, and honest about what we can claim.

What a Future Team Should Do

The Study Ladder

Target: Kirkpatrick Level 2 (Learning). Levels 3–4 require future validation.



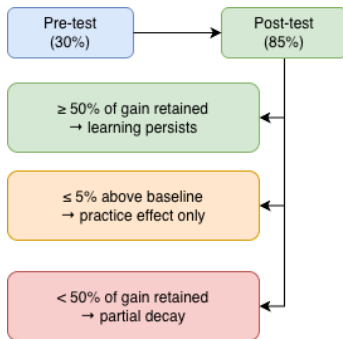
Kirkpatrick Level 2 target — Arthur et al.: learning gain is the standard training evaluation target [10].
Level 3–4 require future validation.

Study 1: Novice Pre/Post Design

Segment	Time	Activity
Consent + intake	3–5 min	Background
Pre-test tray sort	5–8 min	No hints, no feedback
Retrieval cards	10–12 min	Prompt-before-reveal
Quiz	5–8 min	With confidence
Practice sort	8–10 min	Error feedback
Post-test	5–8 min	Parallel variant, no hints
Photo-ID test	5–7 min	Convergent validity

n = 8–12 novices, one 60-min session, within-subject

Study 2: Delayed Retention (The Falsification Criterion)



All three must be met for a retention claim.

Criterion 1 — $\geq 50\%$ of gain retained

Criterion 2 — Accuracy $\geq 5pp$ above baseline

Criterion 3 — $\geq 70\%$ return rate

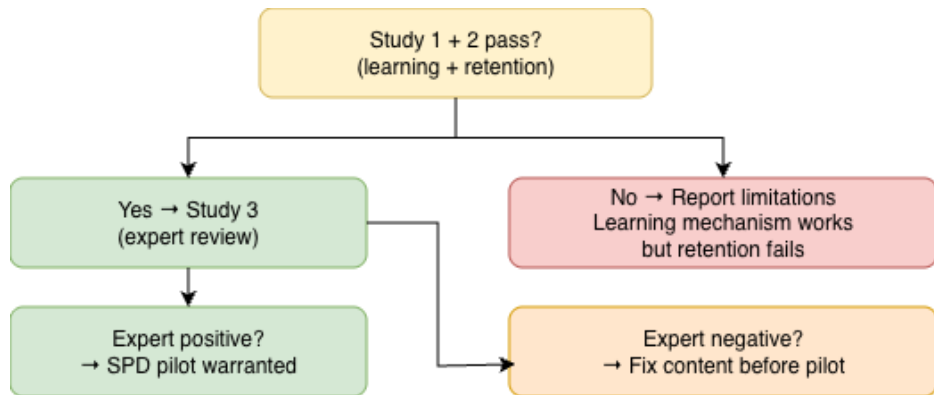
If these fail, the claim reverts to "immediate learning only."

Study 3: Expert Review (Closing the Content Gap)

- Q Are instrument names and aliases correct for local practice?
- Q Are counts and distractors plausible for training?
- Q Are lookalike pairs educationally meaningful?
- Q Are feedback messages misleading or safe?
- Q Would the weak-item report help target coaching?

The main weakness of a student-built module is **content validity**. Expert review closes that gap.

Go/No-Go Decision for Future Team



Study passes → SPD pilot is justified. Study fails → problem may be harder than training alone.

Conclusion

What We Know and What We Don't

What we know

- Tray errors are systemic, recurring, and well-documented
- CV alone can't solve this — the validation burden is too high
- Training is the buildable entry point
- The module schema is durable infrastructure
- The study ladder has falsification criteria

What we can't claim (yet)

- Training transfers to real SPD work
- System works for experienced technicians
- It reduces OR delays
- CV is deployment-ready
- Same-day gains persist (Study 2 tests this)

The Honest Handoff

Our project found that this problem is harder and more systemic than any single technical fix can address. Our contribution is documenting *why*, and giving the next team a validated starting point — not a finished product.

The study is designed. The schema is defined. The scoring is tested. **Run the study.**

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